TASK: 3(A) Array Manipulation And Analysis Programs.

Return The Max sliding Window

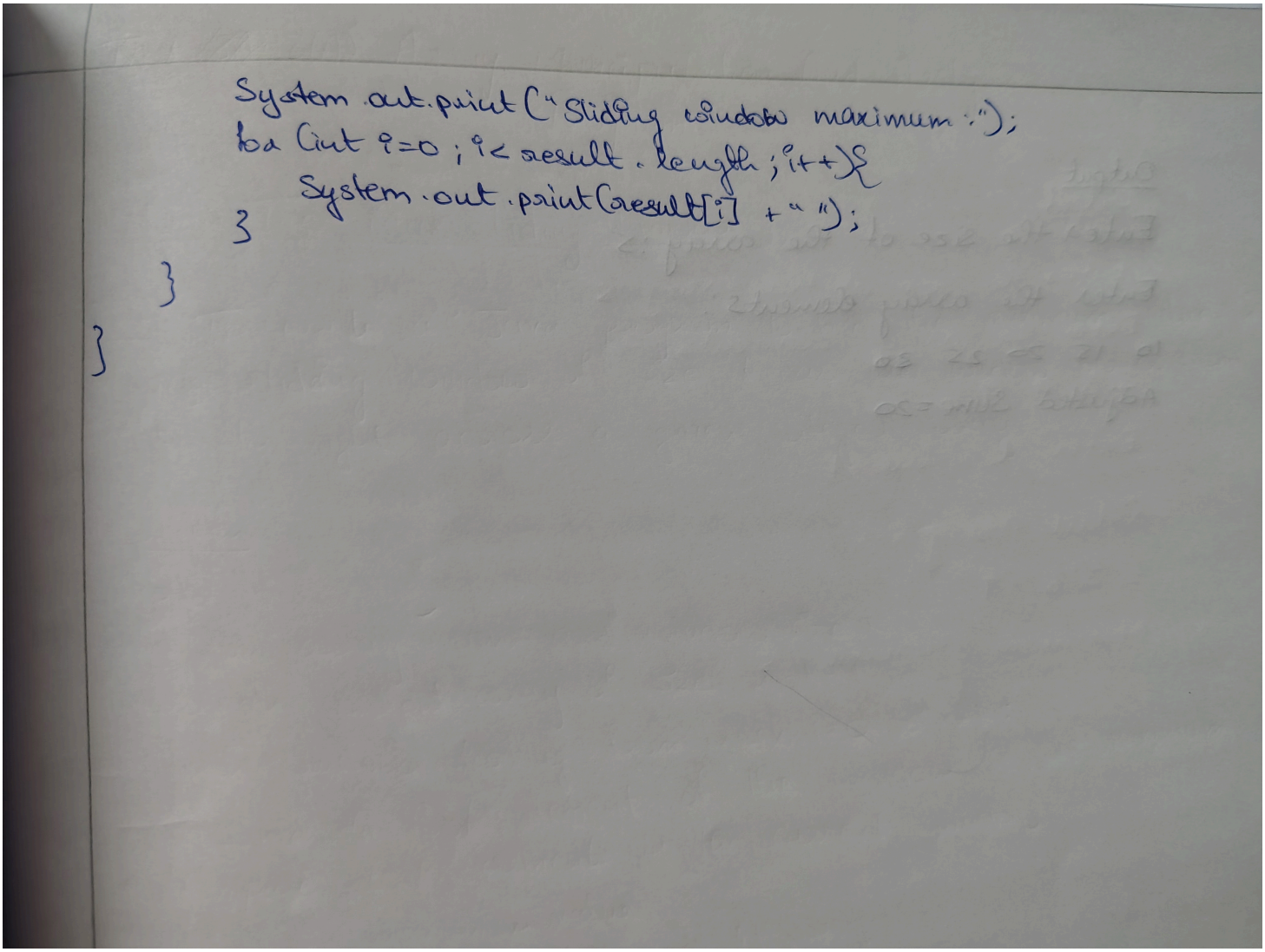
Aim: To write a Java program to find the maximum value in each sliding window of size k as the window moves from left to right across a given integer array.

Algorithm:

(1) Start
(2) Read the array nums and window size k
(3) Create an output array of size n-k+1
(4) For each window starting from index i=0 to n-k:
   * Assume the first element of the window is the maximum
   * Compare all k elements in the current window
   * update the maximum value
   * store the maximum in the result array
(5) Print the result.

Program:

```
import java.util.Scanner;
class SlidingWindowMaximum{
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter array size:");
        int n = sc.nextInt();
        int[] nums = new int[n];
        System.out.println("Enter "+n+" array elements:");
        for(int i=0;i<n;i++){
            nums[i] = sc.nextInt();
        }
        System.out.print("Enter window size:");
        int k = sc.nextInt();
        int[] result = new int[n-k+1];
        for(int i=0;i<=n-k;i++){
            int max = nums[i];
            for(int j=i;j<i+k;j++){
                if(nums[j]>max){
                    max = nums[j];
                }
            }
            result[i] = max;
        }
```



```
        System.out.print("Sliding window maximum:");
        for(int i=0; i< result.length; i++){
            System.out.print(result[i] + " ");
        }
    }
}
```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Output

Enter array size : 8

Enter 8 array elements:

1 3 -1 -3 5 3 6 7

Enter window maximum : 3 3 5 5 6 7

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TASK : 3(B)

Adjusted Sum

Aim: To write a Java program that calculates the adjusted sum of an integer array by adding all even numbers and subtracting all odd numbers.

Algorithm:

(1) Start
(2) Read the value of N
(3) Read N integer elements into the array
(4) Initialize a variable sum to 0.
(5) Traverse each element of the array:
   * if the element is even, add it to sum.
   * If the element is odd, subtract it from sum.
(6) Display the final adjusted sum.
(7) Stop

Program

```
import java.util.Scanner;
class AdjustedSum{
  public static void main(String[] args){
     Scanner sc =new Scanner (System.in);
     System.out.print("Enter the size of the array:");
     int N = sc.nextInt();
     int[] arr = new int[N];
     int sum =0;
     System.out.println("Enter the array elements:");
     for(int i=0;i<N;i++){
        arr[i] = sc.nextInt();
        if (arr[i] %2 ==0){
           sum =sum + arr[i];
        }else{
           sum = sum - arr[i];
        }
     }
     System.out.println("Adjusted Sum =" +sum);
  }
}
```

VEL TECH-CSE

EX NO.	
PERFORMANCE (5)	
RESULT AND ANALYSIS (5)	
VIVA VOCE (5)	

Output

Enter the size of the array : 5

Enter the array elements :

10 15 20 25 30

Adjusted Sum = 20

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TASK-3(C) Rotate array and find Max Difference

Aim: To write a java program that rotates an array to the right by k positions and finds the maximum absolute difference between any two adjacent elements after rotation.

Algorithm:

(1) Start
(2) Read the size of the array N.
3. Read N elements into the array.
4. Read the value of k.
5. Rotate the array to the right by k positions using the formula:
   rotated[(i+k) % N] = arr[i].
6. Traverse the rotated array and calculate the absolute difference between adjacent elements using Math.abs().
7. Store the maximum difference found.
8. Display the rotated array and the maximum difference.
9. Stop.

Program:

```
import java.util.Scanner;
class RotateMaxDifference {
  public static void main(String[] args){
    Scanner sc = new Scanner(System.in);
    System.out.print("Enter the size of the array: ");
    int N = sc.nextInt();
    int[] arr = new int[N];
    int[] rotated = new int[N];
    System.out.println("Enter the array elements: ");
    for (int i = 0; i<N; i++){
      arr[i] = sc.nextInt();
    }
    System.out.print("Enter k value: ");
    int k = sc.nextInt();
    k = k%N;
    for (int i=0; i<N; i++){
      rotated[(i+k)%N] = arr[i];
    }
    int maxDiff = 0;
    for(int i=0; i<N-1; i++){
      int diff = Math.abs(rotated[i]-rotated[i+1]);
      if (diff > maxDiff){
        maxDiff = diff;
      }
    }
    System.out.println("Rotated Array:");
    for(int num : rotated){
      System.out.print(num+" ");
    }
    System.out.println("\nMaximum Absolute Difference: "+maxDiff);
  }
}
```

Output:

```
Enter the size of the array: 5
Enter the array elements:
1 2 3 4 5
Enter k value: 2
Rotated Array:
4 5 1 2 3
Maximum Absolute Difference: 3
```

Result: Thus the program was executed successfully and the output was verified.

Output

Enter the size of the array : 5

Enter the array elements :

3 8 9 7 6

Enter k value : 1

Rotate Array :

6 3 8 9 7

Maximum difference = 5

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```
    }
    System.out.println("\n Maximumdifference =" + maxDiff);
  }
}
```

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EX NO.	3
PERFORMANCE (5)	5
RESULT AND ANALYSIS (5)	5
VIVA VOCE (5)	5
RECORD (5)	5
TOTAL (20)	20
SIGN WITH DATE	10/2/26.

Result: Thus, the java program successfully rotates the array to the right by k positions and finds the maximum absolute difference between adjacent elements.